RECEIPT SCANNER

VOLUME 4

SOFTWARE TESTING

RECEIPT SCANNER

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RECEIPT SCANNER

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CHAPTER 1
Scope

1.1 Introduction

The Receipt Scanner V2 Software Testing Document (STD) defines the testing strategy, test cases, and validation procedures used to verify that the system satisfies the requirements specified in the Software Requirements Specification (SRS).

The Receipt Scanner V2 system processes receipt images using OCR, extracts structured financial data, and provides storage, aggregation, and export functionality. Testing ensures that each stage of this processing pipeline operates correctly, reliably, and in accordance with defined requirements.

1.2 Document Overview

This chapter discusses the scope of the Receipt Scanner V2 Software Testing Document (STD). This document is taken in conjunction with the Software Requirements Description (SRS) to test the requirement fulfillment of the Receipt Scanner V2 system.

This document is organized into five chapters. Chapter one covers the scope of the document and the receipt scanner system. Chapter 2 describes the testing framework and objectives and introduces the receipt scanner v2 testing frameworks. Chapter 3 describes the valid tests. Chapter 4 describes the error tests.

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CHAPTER 2
Testing Framework

2.1 Introduction

This chapter describes the testing objectives and approach for the receipt scanner v2 system. This chapter describes the testing objectives, approach, and categories, as well as the methods of test verification and the testing environment.

2.2 Testing Objectives

The objectives of testing are to:

- Verify that all functional requirements are correctly implemented
- Validate correct system behavior under expected inputs
- Ensure the system handles invalid or unexpected inputs safely
- Confirm data integrity across processing, storage, and export
- Validate user interaction flows and usability

2.3 Testing Approach

Testing follows a black-box functional testing model supplemented by component-aware validation based on the system pipeline:

From SWDD pipeline:

Input Validation → Preprocessing → OCR → Parsing → Storage → Aggregation → Export

Each stage is tested independently and in integration.

Each test case will contain the following components:

- Test Description
- Test Methodology
- Expected Results
- Actual Results

Each category will have the following traits:

- Test Description
  - Test Status (New, Modified, Deleted)
- Test Methodology
  - Externa File Location (input files if applicable)
- Expected Results
  - Test Status (New, Modified, Re-used, Deleted)
- Actual Results
  - Execution Date
  - Execution Status
  - External File Location (output files if applicable)

2.4 Test Categories

Tests are divided into three primary classes:

Table 2-1 – Test Categories

CATEGORY	DESCRIPTION
Valid Tests	Confirm the correct operation under expected conditions
Error Tests	Validate handling of invalid inputs or failures
Interaction Tests	Validate UI behavior and user workflows

2.5 Test Environment

Testing is performed under:

- Python runtime environment
- Flask-based web interface
- OCR engine (Tesseract or equivalent)
- Local file system for uploads and storage

Constraints reflect SRS operating assumptions .

2.6 Verification Methods

The following verification methods are used.

Table 2-2 – Verification Methods

METHOD	DESCRIPTION
Test (T)	Execute system with defined inputs
Inspection (I)	Review outputs without execution
Demonstration (D)	Observe system behavior interactively

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CHAPTER 3
Valid Test Cases

3.1 Introduction

This chapter discusses the valid test cases of Receipt Scanner V2

3.2 Test Cases

3.2.1 *Suite 01*

Table 3-1 - Suite 01 - Receipt Upload and Validation

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V1				
Test Description	New			
This test verifies the capability to upload a single receipt.				
Test Methodology				
Select Gas type → upload **Untitled.png** → click Upload Receipts.				Suite01/data/Untitled.png
Expected Results	New			
The receipt is accepted.				
Actual Results				
A new record appeared with date 2024-01-02 and missing total.		4/19/2026	Pass	

Table 3-1 - Suite 01 - Receipt Upload and Validation

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V2				
Test Description	New			
This test verifies the capability to upload multiple receipts.				
Test Methodology				
Select appropriate receipt types → upload **Food_Blurry.jpg**, **Fake2.jpg**, **Lowes.jpg**, and **Sephora.jpg** sequentially → click Upload Receipts for each.				Suite01/data/Food_Blurry.jpg, Suite01/data/Fake2.jpg, Suite01/data/Lowes.jpg, Suite01/data/Sephora.jpg
Expected Results	New			
All receipts are accepted and stored				
Actual Results				
Each uploaded receipt created a new record in the system and was stored successfully.		4/19/2026	Pass	

Table 3-1 - Suite 01 - Receipt Upload and Validation

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V3				
Test Description	New			
This test verifies that valid file formats are accepted. This test has two executions, one with a PNG file and one with a JPG file.				
Test Methodology				
Execution 1: Upload **Untitled.png** Execution 2: Upload **Fake2.jpg** Execution 3: Upload **lowes.pdf** (invalid control test)				Suite01/data/Untitled.png, Suite01/data/Fake2.jpg, Suite01/data/lowes.pdf
Expected Results	New			
Both executions are accepted/uploaded.				
Actual Results				
PNG and JPG files were successfully uploaded. The PDF file was rejected with an error message indicating unsupported file type.		4/19/2026	Pass	

3.2.2 *Suite 02*

Table 3-2 - Suite 02 - OCR and Parsing

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V4				
Test Description	New			
This test verifies that OCR extraction is functioning.				
Test Methodology				
Upload multiple receipts (**Food_Blurry.jpg**, **Fake2.jpg**, **Sephora.jpg**, **Lowes.jpg**) and observe extracted text fields.				Suite02/data/Food_Blurry.jpg, Suite02/data/Fake2.jpg, Suite02/data/Sephora.jpg, Suite02/data/Lowes.jpg
Expected Results	New			
The text from the input file is extracted.				
Actual Results				
OCR successfully extracted text from all images. Some fields (e.g., total in Untitled.png) were not detected due to image quality.		4/19/2026	Pass	

Table 3-2 - Suite 02 - OCR and Parsing

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V5				
Test Description	New			
This test verifies that the parsing capability is functioning.				
Test Methodology				
Upload receipts and observe whether vendor, date, and totals are identified.				Suite02/data/Food_Blurry.jpg, Suite02/data/Fake2.jpg, Suite02/data/Sephora.jpg, Suite02/data/Lowes.jpg
Expected Results	New			
The fields on the input file are properly identified.				
Actual Results				
Fields were identified for valid receipts. Missing fields were flagged with “Needs Review.”		4/19/2026	Pass	

Table 3-2 - Suite 02 - OCR and Parsing

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V6				
Test Description	New			
This test verifies that the parsing capability is functioning.				
Test Methodology				
Upload receipts and verify extracted data is populated in structured format.				Suite02/data/Food_Blurry.jpg, Suite02/data/Fake2.jpg, Suite02/data/Sephora.jpg, Suite02/data/Lowes.jpg
Expected Results	New			
The fields on the input file are properly identified and extracted.				
Actual Results				
Fields were extracted correctly for valid receipts. Incomplete data was flagged appropriately.		4/19/2026	Pass	

3.2.3 *Suite 03*

Table 3-3 - Suite 03 – Data Management

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V7				
Test Description	New			
This test verifies the receipt edit capability is functional.				
Test Methodology				
Open **Sephora.jpg** → edit total from 74.20 to 74.50 → click Save.				
Expected Results	New			
The receipt data is edited.				
Actual Results				
The receipt total was updated to 74.50 and persisted in the system.		4/19/2026	Pass	

Table 3-3 - Suite 03 – Data Management

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V8				
Test Description	New			
This test verifies that data is persistent and retrievable.				
Test Methodology				
Refresh page and navigate away after editing receipts.				
Expected Results	New			
The receipt data persists between sessions and is properly retained.				
Actual Results				
All receipt data remained after refresh and navigation.		4/19/2026	Pass	

3.2.4 *Suite 04*

Table 3-4 - Suite 04 – Aggregation and Summary

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V9				
Test Description	New			
This test verifies that multiple receipts are shown properly in the monthly summary.				
Test Methodology				
Upload multiple receipts → view Monthly Summary panel.				Suite04/data/Sephora.jpg, Suite04/data/Lowes.jpg
Expected Results	New			
The receipts are grouped properly /				
Actual Results				
Receipts were grouped by month (e.g., 2026-04, 2026-10).		4/19/2026	Pass	

Table 3-4 - Suite 04 – Aggregation and Summary

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V10				
Test Description	New			
This test verifies that the totals in the monthly summary are correct.				
Test Methodology				
Compare calculated totals with known receipt values.				
Expected Results	New			
The receipts in the group show the correct total in the monthly summary.				
Actual Results				
Totals matched expected sums (e.g., April = $86.15).		4/19/2026	Pass	

Table 3-4 - Suite 04 – Aggregation and Summary

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V11				
Test Description	New			
This test verifies that receipts that are excluded from the summary via the toggle summary button are not included in the aggregate report.				
Test Methodology				
Click Toggle Summary on a valid receipt.				
Expected Results	New			
The receipts totals are correct.				
Actual Results				
Receipt was removed from summary and totals updated correctly.		4/19/2026	Pass	

3.2.5 *Suite 05*

Table 3-5 - Suite 05 – Export

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V12				
Test Description	New			
This test verifies the export capability is functional. This test has two executions: The first execution contains only valid receipts. The second execution contains a mix of valid, invalid, and broken receipts.				
Test Methodology				
Click Export Current View with mixed receipts.				
Expected Results	New			
A valid CSV file is produced and the data in the file contains the correct information. For the second execution, invalid receipts are included in the export, but broken receipts are not.				
Actual Results				
CSV file downloaded successfully. Data generation executed without error.		4/19/2026	Pass	Suite05/TCV12/output.csv

3.2.6 *Suite 06*

Table 3-6 - Suite 06 – UI Interaction

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V13				
Test Description	New			
This test verifies that receipts are displayed correctly in the edit menu.				
Test Methodology				
Open receipt in edit mode.				
Expected Results	New			
Both preprocessed and post-processed receipts are shown.				
Actual Results				
Both original and processed images displayed correctly.		4/19/2026	Pass	

Table 3-6 - Suite 06 – UI Interaction

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V14				
Test Description	New			
This test verifies that status messages and receipt status are shown correctly. This test has two executions. The first involves uploading an invalid receipt type (PDF) and seeing the status message. The second involves editing and saving a valid receipt and observing the status message.				
Test Methodology				
Execution 1: Upload **lowes.pdf** Execution 2: Edit and save **Sephora.jpg**				Suite06/data/lowes.pdf, Suite06/data/Sephora.jpg
Expected Results	New			
The expected status messages for each execution is displayed.				
Actual Results				
Error message shown for PDF upload. Success message shown after edit.		4/19/2026	Pass	

Table 3-6 - Suite 06 – UI Interaction

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V15				
Test Description	New			
This test verifies the ability to select a receipt type prior to uploading				
Test Methodology				
Select type → upload receipts.				Suite06/data/Sephora.jpg, Suite06/data/Fake2.jpg, Suite06/data/Lowes.jpg
Expected Results	New			
The selection is able to be made and after upload, the receipt is categorized properly.				
Actual Results				
Receipts were assigned correct categories (Gas, Food, Retail).		4/19/2026	Pass	

Table 3-6 - Suite 06 – UI Interaction

DESCRIPTION	TEST STATUS	EXECUTION DATE	EXECUTION STATUS	EXTERNAL FILES
TC-V16				
Test Description				
This test verifies the ability to exclude a receipt from a summary (interaction)				
Test Methodology	New			
Click Toggle Summary on a valid receipt.				
Expected Results	New			
The selection is able to be excluded with the button.				
Actual Results				
Receipt removed from summary and totals updated.		4/19/2026	Pass	

APPENDIX A

Nomenclature

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NOMENCLATURE

Aggregation	The process of computing grouped financial data such as monthly totals
API	Application Programming Interface
Category	A classification applied to a receipt (e.g., Gas, Food, Retail)
Collection	A grouping of receipts within the system
CSV	Comma-Separated Values
Export	The generation of external data files (e.g., CSV) from stored receipt data
High Total Flag	A validation flag triggered when a detected total exceeds expected thresholds
Invalid Receipt	A receipt that fails validation checks but is still processed
Missing Field	A required data element (e.g., total or date) not extracted
Needs Review	A system status indicating incomplete or uncertain extraction
OCR	Optical Character Recognition
OCR Extraction	The process of converting image-based text into machine-readable text
Parsing	The process of interpreting OCR output to identify structured data fields
Preprocessing	Image transformations applied prior to OCR (e.g., grayscale, thresholding)
Receipt	An image or document representing a transaction used as system input

NOMENCLATURE

Receipt Scanner V2	The system under test that processes receipt images and extracts financial data
RTM	Requirements Traceability Matrix
SRS	Software Requirements Specification
STD	Software Testing Document
Storage	Persistence of receipt data within the system
Structured Data	Extracted fields such as vendor, date, and total
SWDD	Software Design Description
Test Case (TC)	A defined set of inputs, conditions, and expected results used to validate system behavior
Test Methodology	The procedure followed to execute a test case
Test Suite	A collection of related test cases grouped by functionality
UI	User Interface
Upload	The process of submitting receipt files to the system
Valid Receipt	A receipt with all required fields successfully extracted

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